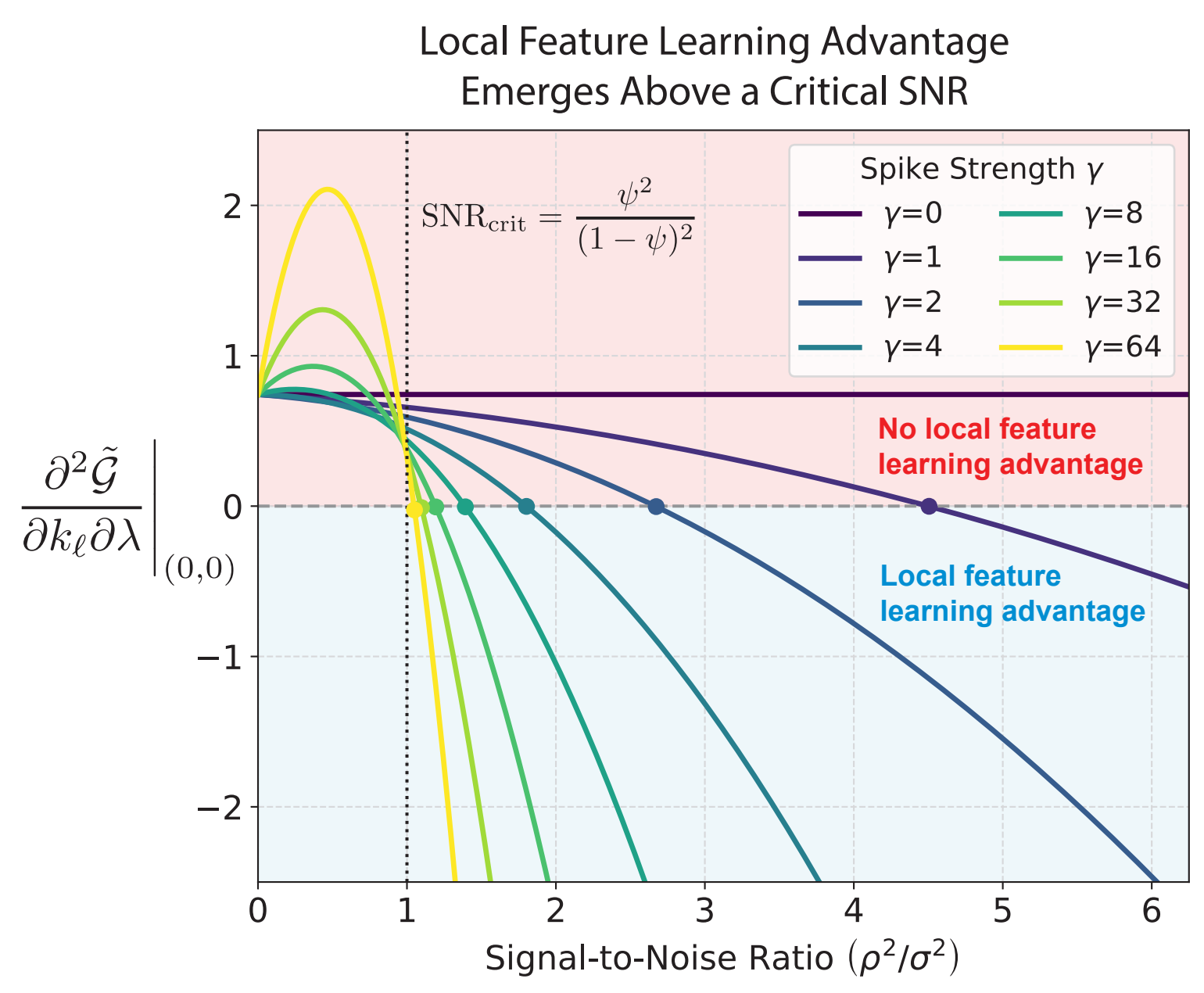

Local Feature Learning Advantage
Emerges Above a Critical SNR
$\mathrm{SNR}_{\mathrm{crit}} = \frac{\psi^2}{(1-\psi)^2}$
Spike Strength $\gamma$
$\gamma$=0
$\gamma$=1
$\gamma$=2
$\gamma$=4
$\gamma$=8
$\gamma$=16
$\gamma$=32
$\gamma$=64
No local feature learning advantage
Local feature learning advantage
$\left.\frac{\partial^2 \tilde{\mathcal{G}}}{\partial k_\ell \partial \lambda}\right|_{(0,0)}$
2
1
0
−1
−2
0
1
2
3
4
5
6
Signal-to-Noise Ratio $(\rho^2/\sigma^2)$